Name:_____

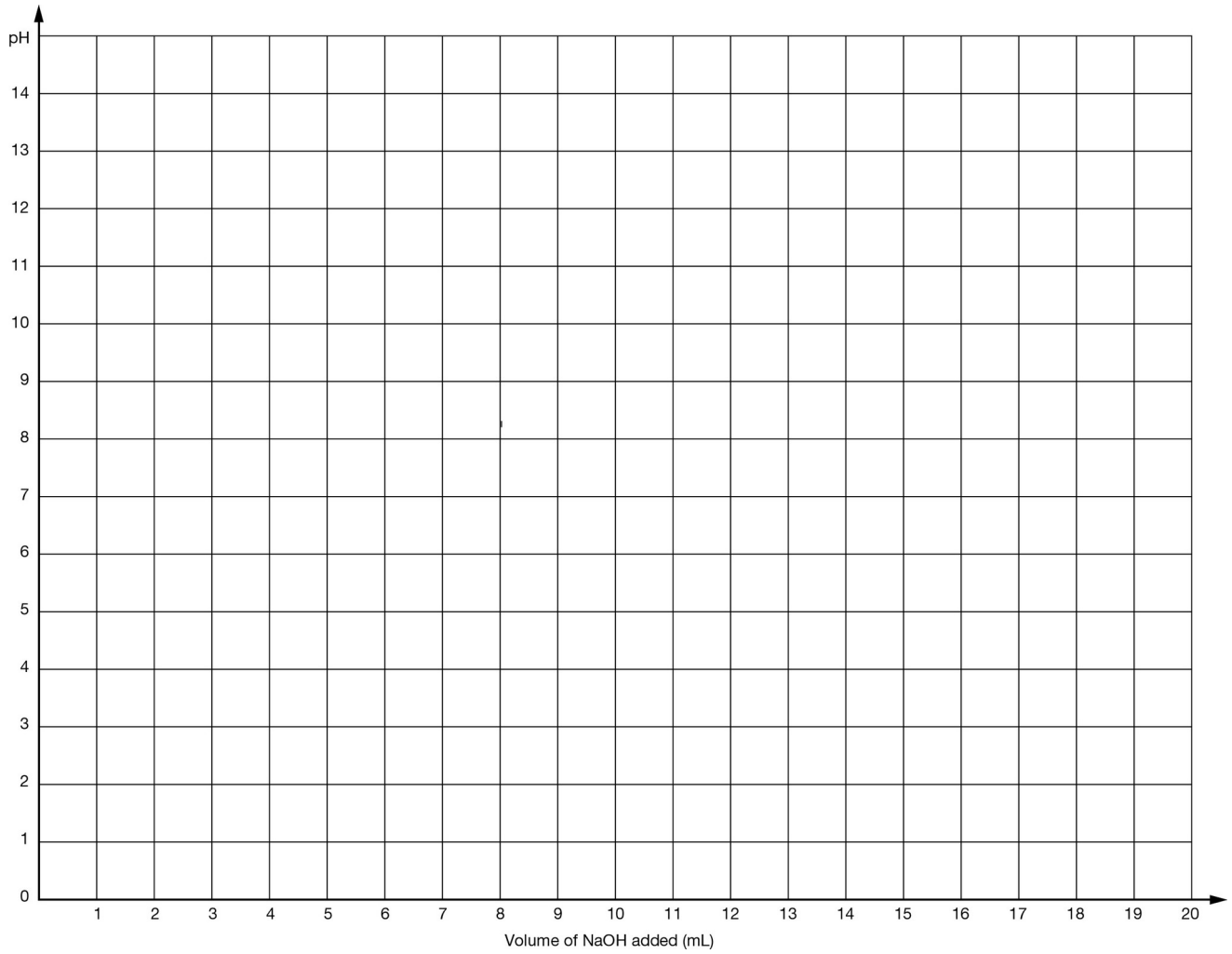
Plotting an acid/base reaction

SCIENCE UNDERSTANDING VISUAL/SPATIAL LOGICAL/MATHEMATICAL

A solution of sodium hydroxide was slowly added to a solution of hydrochloric acid. As the sodium hydroxide was added, the pH of the solution was continually measured using a pH meter. The results are shown in the table below:

Volume of NaOH added (mL)	pH	Volume of NaOH added (mL)	pH
0	2.5	12.5	8.5
2	3.2	13	11.6
4	3.6	14	12.2
6	4.0	15	12.4
8	4.2	17	12.8
10	4.6	19	13.0
12	5.4	20	13.1

1 Use this data to construct a **line graph** that shows what happened to the pH as sodium hydroxide was added. Connect up the points with a smooth curve.


pH
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
Volume of NaOH added (mL)

2 Classify the type of reaction happening here. Was it an acid/metal, neutralisation or acid/carbonate reaction?

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3 State what the pH of the hydrochloric acid solution was before any reaction had taken place.

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4 State the pH of the solution at the end of the reaction.

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5 State whether the solution was acidic, basic or neutral when the following volumes of sodium hydroxide had been added:

a 5 mL .....

b 16 mL .....

c 19 mL .....

d 10 mL .....

6 State the approximate volume of sodium hydroxide that needed to be added to exactly neutralise the acid (when the pH is approximately 7).

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